

Transformer Architecture Notes

Attention Is All You Need: Scaled Dot-Product & Feed-Forward Networks

Core Architecture Equations

Self-Attention: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$

Multi-Head: $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$

FFN: $\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$

Section 1: Basic Level

Question 1: Scaled Dot-Product Calculation

Query vector $\mathbf{Q} = [1, 2]$, Key vector $\mathbf{K} = [2, 1]$, and dimension $d_k = 2$. Calculate the scaled score before softmax.

Step-by-Step Solution:

1. Dot Product (QK^T): $(1*2) + (2*1) = 4$

2. Scaling Factor: $\sqrt{d_k} = \sqrt{2} \approx 1.414$

3. Final Calculation: $4 / 1.414 \approx 2.828$

Answer: Scaled Score ≈ 2.83

Question 2: Multi-Head Dimensions

If $d_{\text{model}} = 512$ and number of heads $h = 8$, find the dimension of each head (d_k).

Step-by-Step Solution:

1. Formula: $d_k = d_{\text{model}} / h$

2. Calculation: $512 / 8 = 64$

Answer: Each head dimension = 64

Section 2: Advanced Level

Question 3: Attention Weight Distribution

Scaled scores for a word attending to itself and one other word are $[2.0, 0.5]$. Calculate the softmax attention weights.

Step-by-Step Solution:

1. Exponentials: $e^{2.0} \approx 7.389$, $e^{0.5} \approx 1.649$

2. Sum: $7.389 + 1.649 = 9.038$

3. Normalize: $W1 = 7.389/9.038 \approx 0.8175$, $W2 = 1.649/9.038 \approx 0.1825$

Answer: Weights $\approx [0.82, 0.18]$

Question 4: FFN Parameter Count

Calculate total weights for a Feed-Forward Network where $d_{\text{model}} = 512$ and $d_{\text{ff}} = 2048$.

Step-by-Step Solution:

1. Layer 1 (512 to 2048): $512 * 2048 = 1,048,576$

2. Layer 2 (2048 to 512): $2048 * 512 = 1,048,576$

3. Total: $1,048,576 + 1,048,576 = 2,097,152$

Answer: Total weights = 2,097,152 (approx. 2.1M)